

Ideation Phase - Brainstorm & Idea Prioritization

Parameter	Detail
Date	18 June 2026
Team ID	Book-a-Doctor Platform
Project Name	Book a Doctor
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Focus Statement

"How might we design a secure, highly responsive, and reliable digital platform that bridges the communication and coordination gap between patients, medical practitioners, and system administrators to facilitate seamless scheduling and clinical data audits?"

Core Selected Problem Statement

Patients struggle with double-bookings, unverified practitioner credentials, and insecure medical record sharing, while doctors lack automated scheduling logs and digital prescription trails. System administrators require a centralized dashboard to vet medical practitioners and ensure platform integrity.

Step-2: Brainstorm, Idea Listing and Grouping

1. User Authentications & Authorization (Security Group)

- Role-Based Access Control (RBAC) separating Patient, Doctor, and Admin views.

- Dual-token JWT validation (Access Token + HttpOnly Cookies Refresh Token) to prevent hijack vectors.
- Password hashing using bcrypt with high work factor (salt rounds: 10).

2. Practitioner Credential Vetting (Trust Group)

- A "Pending" registration state for newly registered doctors.
- An administrator vetting dashboard to inspect experience, specialty, and hospital certificates.
- One-click approval/rejection toggle which instantly syncs with patient-facing search directories.

3. Smart Consultation Scheduling (Scheduling Group)

- Double-booking validation engine on the backend (Mongoose transactional checks).
- Automatic notification alerts upon status changes (Pending → Approved/Rejected).
- Doctor-controlled slots manager.

4. Digital Medical Locker (Data Storage Group)

- Patient-owned PDF/Image upload system utilizing Multer middleware.
- File validation rules restricting MIME types to PDF/JPEG/PNG.
- Shared access control letting assigned doctors view locker records during active appointment timelines.

Step-3: Idea Prioritization

High Priority (Sprint 1 & 2 Core)

- JWT Dual-Token authentication system.
- Doctor registration and Admin vetting queue.
- Appointment scheduling calendar logic with double-booking prevention.

Medium Priority (Sprint 3 Core)

- Digital medical locker (Multer file upload + file validation rules).
- In-app notification logging service.
- Interactive dashboards with light/dark theme toggles.

Low Priority (Future Roadmaps)

- Real-time video conferencing (telehealth consultation) integrations.
- Third-party payment gateway integration (Stripe/Razorpay) for premium slot booking.